



Sodium Flowers

An Overview

⋮ MEL Science

Experiment

**How to grow
liquid flowers**

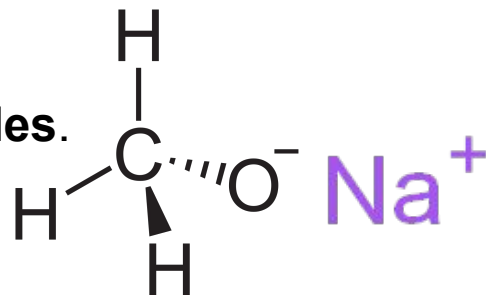


Method:

1. Pour **isopropanol** and a little **ethanol** into the **beaker** and stir the solution.
2. Sprinkle the **bromothymol blue** and **phenolphthalein indicators** into the solution.
3. Pour the resulting indicator mixture into a **glass bowl** and add a few pieces of **metallic sodium** on top.
4. Watch as blue-red patterns that look like flowers appear on the surface of the solution.

How does it Work?

- The Na reacts with the alcohol $\rightarrow$ forming **sodium alkoxides**.



- The solution will have an **alkaline** medium $\rightarrow \text{pH} > 7$
- The indicators react to this change in pH.

- This turns the mixture blue and crimson.

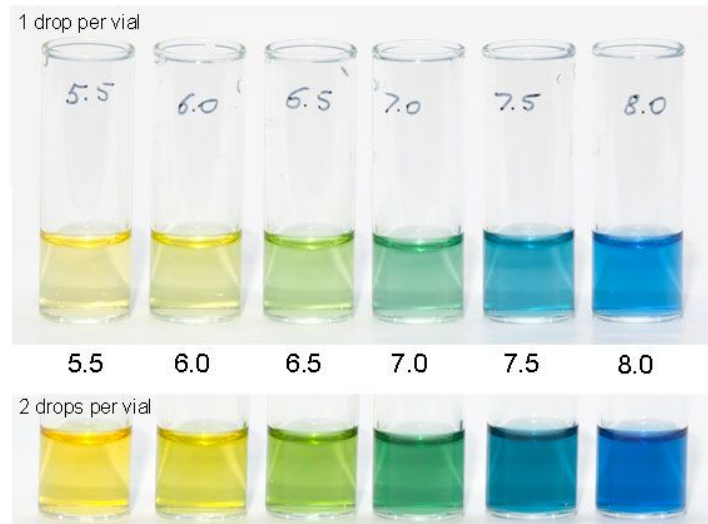
Bromothymol blue $\rightarrow$ blue.

Phenolphthalein $\rightarrow$ pink.



Pink
pH 8.2-12

Bromothymol Blue pH Tester
pH Color Chart



How does it Work?

- As isopropanol has quite a **high viscosity**.
 - Viscosity = how easily it flows.
 - Low viscosity = flows easily.
 - High viscosity = does not flow easily.
- The **slow diffusion** of the **alkali** takes place.
- Makes the flower appear.

